# Esophageal Cancer Treatment

### Esophageal Cancer Treatment Categories

| Category | AJCC | Stage | Treatment |
|---|---|---|---|
| High-grade Dyplasia | 0 | Tis | Radiofrequency Ablation |
| Superficial Tumors | I | T1a | Endoscopic Therapy |
| Localized Tumors | IIB | T1b T2 | Surgery |
| Locally-advanced | III IIA | T3 or $N^+$ | Chemo $\pm$ RT $\rightarrow$ Surgery |
| Extra-Regional | IVA | N3 | Chemo$\rightarrow$ ChemoRT$\rightarrow$ Surgery |
| Metastatic | IVB | M1 | Chemotherapy $\pm$Radiation |

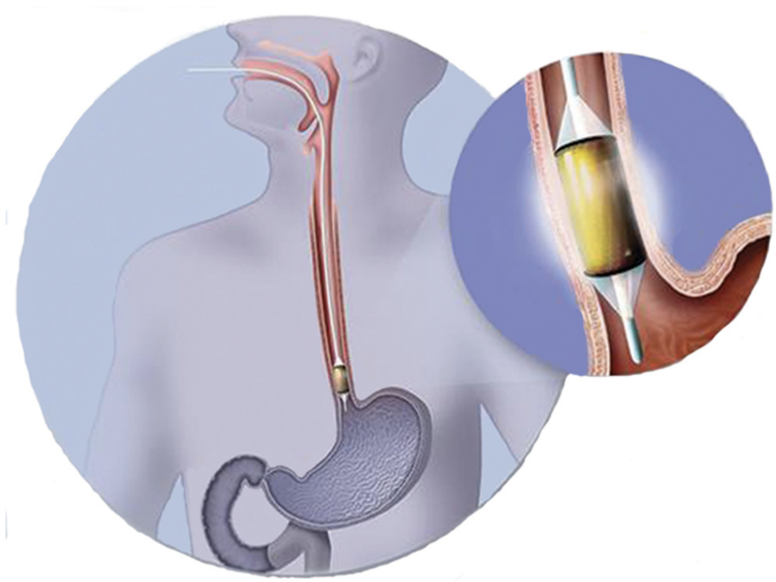
### High-grade Dyplasia => Radiofrequency Ablation

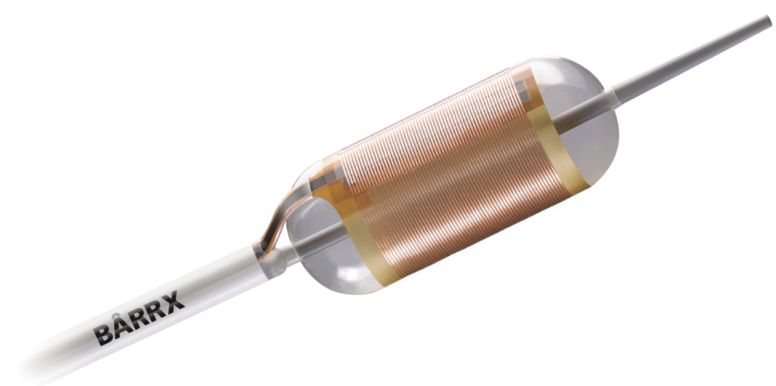
127 patients with dysplasia randomized:

- Radio-frequency ablation
- Sham ablation

Low-grade dysplasia in 64

High-grade dysplasia in 63

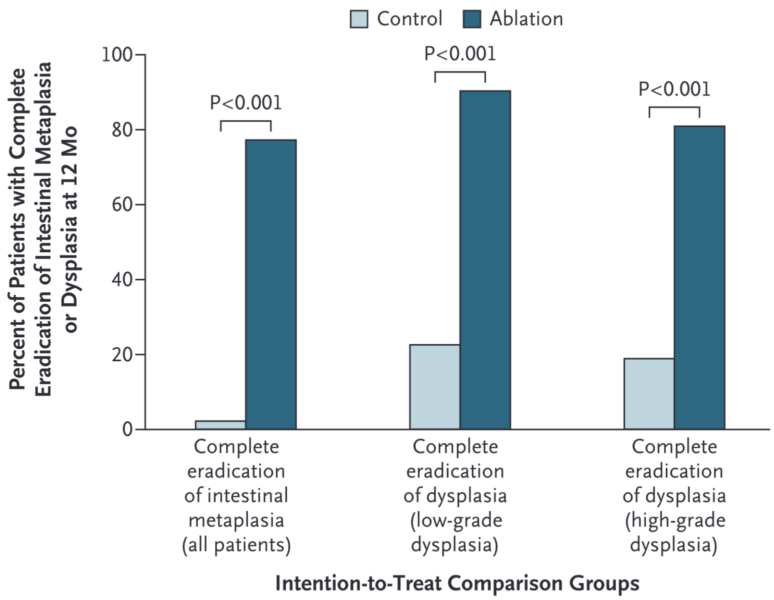





(Shaheen et al. 2009)

## Radiofrequency Ablation for Dysplasia

Radiofrequency Ablation results in eradication of Barrett's in 75% at 1 year




No role for primary surgery in high-grade dysplasia

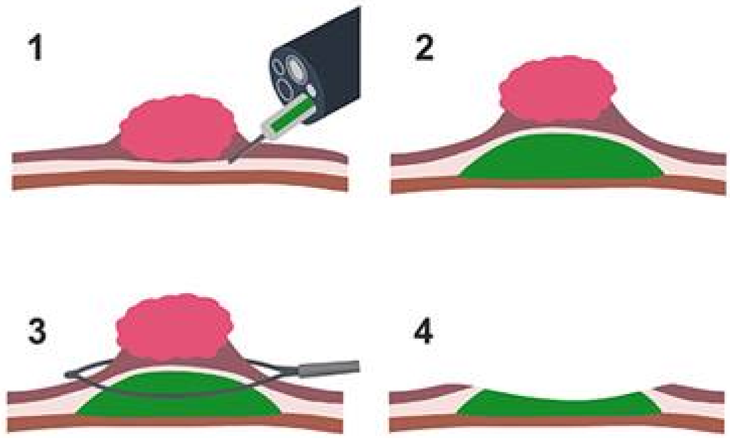
(Shaheen et al. 2009)

## Superficial Tumors (T1)

Workup of nodular Barretts:

- Endoscopic Ultrasound
- Endoscopic Mucosal Resection
    - Diagnostic (T staging)
    - May be therapeutic for T1a tumors

## Endoscopic Musocal Resection for T1 Tumors




## Endoscopic Submucosal Dissection for T1a/T1b

Endoscopic resection uses needle-knife to dissect *below* submucosa

Maybe suitable for T1b lesions *if* lesion is completely resected

Risk of perforation

## Localized Tumors (T2 N0)

Patients staged as uT2 N0 are candidates for primary surgery. *However:*

- EUS has a 25% rate of understaging uT2 N0 tumors
- Understaged patients who undergo primary surgery would need chemo or chemoRT postop

### Asymptomatic Tumors (minimal dysphagia)

- EUS to distinguish T2 from T3 tumors
- If uT2 N0 → CT chest/abdomen/pelvis → Esophagectomy
- If uT3 or N1 → PET → neoadjuvant therapy

Patients with dysphagia almost always are T3 tumors (and don't need EUS)

### Symptomatic Tumors (dysphagia)

Patients with dysphagia to solids or weight loss or tumor length >3cm are unlikely to have T1-2 tumors and can be initially evaluated with PET Scan

- Disease confined to the esophagus and regional nodes → Locally-advanced
- Metastatic disease → Metastatic
- N3 → Extra-Regional

### EUS in Patients with Dysphagia

Memorial Sloan Kettering patients with esophageal cancer:

- 61 with dysphagia, 54 (89%) were found on EUS to have uT3-4 tumors.
- 53 without dysphagia, 25 (47%) were uT1-2 → candidates for primary surgery.

EUS can be omitted for patients with dysphagia

EUS can useful in patients *without* dysphagia (to identify T2N0 → Surgery)

(Ripley et al. 2016)

### PET Scan

PET has more specificity and sensitivity than CT in detecting regional lymph node and distal metastasis

## Locally-advanced

Improved survival with adjunctive therapy. There are two options:

- ChemoRT → Surgery (CROSS Trial)
- Chemo → Surgery → Chemo (EsoPEC Trial)

## CROSS Surgery vs ChemoRT →Surgery

368 esophageal cancer patients randomized:

- Surgery alone
- Chemo+RT → Surgery

75% adenocarcinoma. T3: 80% T2: 17%. age $\tilde{x}$=60

⇒ longer survival with Chemo+RT → Surgery

Chemotherapy: Weekly carboplatin and paclitaxel

Radiation: 4140 cGy in 23 fractions (180cGy/fraction)

(Shapiro et al. 2015)

## CROSS Surgery vs ChemoRT →Surgery

| | Surgery | CRT→Surgery | |
|---|---|---|---|
| Time to Surgery | 24d | 97d | |
| Unresectable | 13% | 4% | p=0.002 |
| R0 Resection | 92% | 69% | p<0.001 |
| Positive Nodes | 75% | 31% | |
| Complications | ~ | ~ | |
| Mortality | ~ | ~ | |

## CROSS - Overall Survival

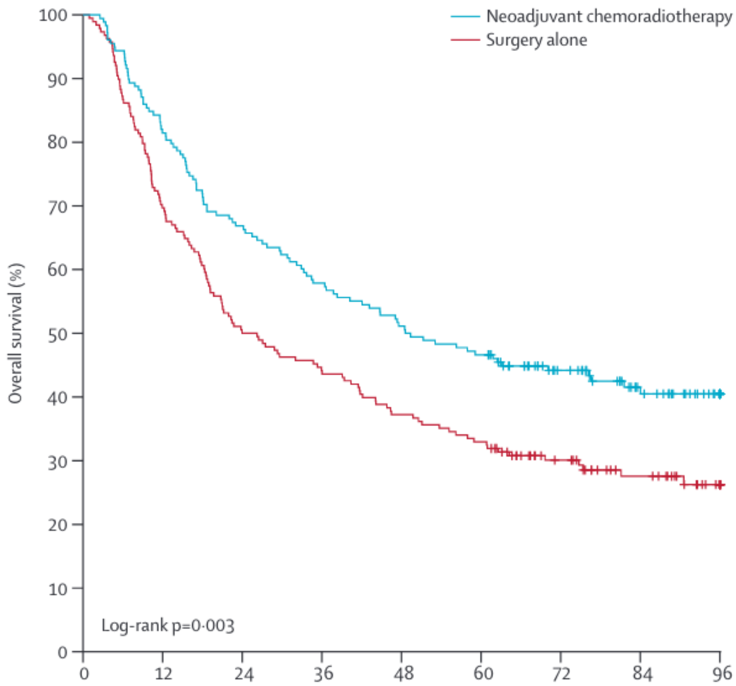



Figure 1: Surgery vs ChemoRT → Surgery

(Shapiro et al. 2015)

## CROSS - Survival by Histology

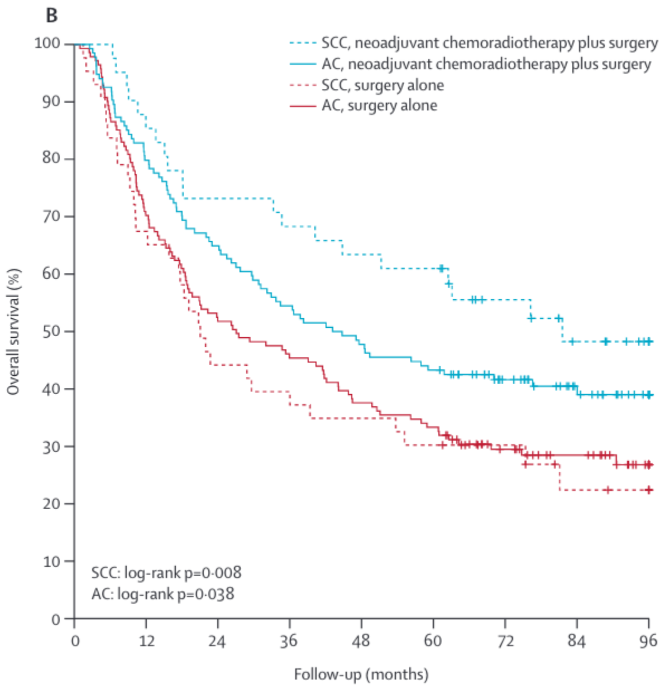



Figure 2: Surgery vs ChemoRT → Surgery

## CROSS Effects by Histology

(Shapiro et al. 2015)

Adenocarcinoma

- Median survival 43mo vs 27mo
- Pathologic complete response in 23%

Squmous Cell Carcinoma

- Median survival 82mo vs 21mo
- Pathologic complete response in 49%

Surgical Therapy of Squamous Cell

(Shapiro et al. 2015)

## Adjuvant Nivolumab after CROSS: Checkmate 577 Trial

EsoCA patients who received ChemoRT$\rightarrow$ Surgery with residual disease (not pCR)

Randomized to one year of nivolumab vs Observation

$\Rightarrow$ Adjuvant nivolumab group had longer median survival: 22mo vs 11mo

(Kelly et al. 2021)

## Checkmate 577 Trial

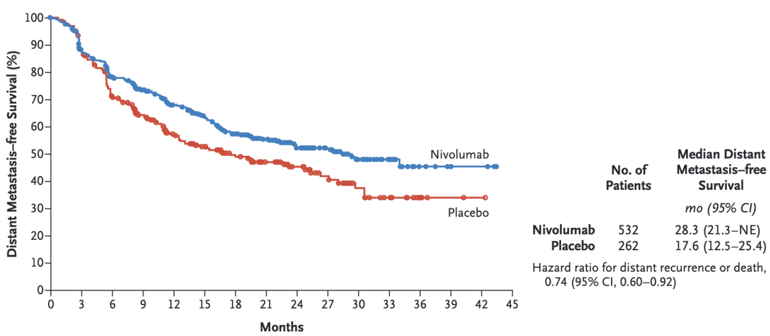



Figure 3: Adjuvant Nivolumab vs Observation

## Nivolumab = Immune Checkpoint Inhibitor (ICI)

(Kelly et al. 2021)

PD-L1 agonist ligand

Interferes with tumor cell down-regulation of T cells

Active against stage IV esophageal cancer

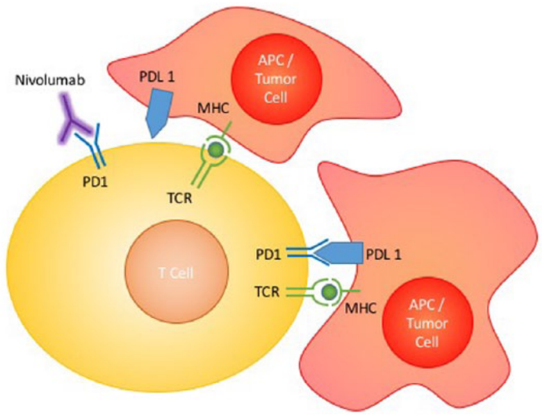



Figure 4: Nivolumab mechanism of action

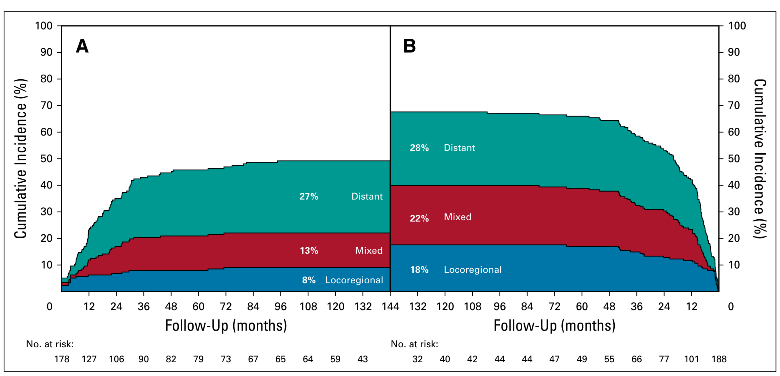
## CROSS - ChemoRT Does Not Reduce Distant Failure

Sites of failure over time

ChemoRT + Surgery *vs* Surgery

ChemoRT appears to reduce risk of local or local+distant failure, but not isolated distant failure

⇒ Can chemotherapy reduce risk of distant failure?




(Shapiro et al. 2015)

## EsoPEC CROSS vs FLOT for Adenocarcionoma

**Adenocarcinoma** esophagus and GE junction

- T1 N+ or T2-4a M0. Median age =63. 89% men

Randomized to CROSS (n=217) vs FLOT chemotherapy (n=221)

- CROSS: carboplatin/paclitaxel + 4140cGy → Surgery
- FLOT: FLOT → Surgery → FLOT

Excluded: Squamous cell, gastric cancer, T1N0, T4b, M1

(Hoeppner et al. 2025)

## EsoPEC: FLOT superior to CROSS

| | FLOT Periop Chemo | CROSS ChemoRT | CROSS Trial ChemoRT |
|---|---|---|---|
| 90d Mortality | 3.2% | 5.6% | |
| $\tilde{x}$ Survival | 66mo | 37mo | 43mo |
| 3year Survival | 57% | 51% | 58% |
| 5year Survival | 51% | 29% | 47% |
| path CR | 17% | 10% | 23% |

See also Neoadjuvant Chemo

(Hoeppner et al. 2025)

## Surgery for Squamous Cell Carcinoma

Squamous Cell Carcinoma of the esophagus

- responds well to chemo+RT
- more difficult to get a surgical margin on the airway
- additional benefit of surgery on top of chemoRT is uncertain

## Squamous Cell: ChemoRT $\pm$ Surgery (FFCD)

Squamous cell of the esophagus $\rightarrow$ 4500cGy RT + 2 cycles of cisplatin + 5FU

Patients with a clinical response were randomized:

- Surgery $\Rightarrow$ 2 year survival 34% Median 17.7mo
- 3 cycles of chemo + 2000 cGy RT $\Rightarrow$ 2 year survival 40% Median 19.3mo

$\Rrightarrow$ No difference in overall survival

(Bedenne et al. 2007)

## Squamous Cell: ChemoRT $\pm$ Surgery (German)

Squamous cell of the esophagus randomized:

- 4000 cGY RT + Chemo $\rightarrow$ Surgery. 64% 2-year PFS. Mortality 12.8%
- 6500cGy RT + Chemo: 41% 2-year PFS. Mortality 3.5%

$\Rrightarrow$ No difference in overall survival

(Stahl et al. 2005)

## Non-Operative Management

## Extra-regional Disease = N3

Patients with N3 disease are Stage IVA and have a poor prognosis.

Treatment consists of chemotherapy followed by radiation.

Patients with persistent disease in the esophagus (and no distant disease) would then be candidates for surgery

### Metastatic = IVB

FOLFOX is first-line systemic therapy for metastatic GI cancers

- Dose-limiting toxicity is frequently peripheral neuropathy

### Adjuvant Immunotherapy: Checkmate 577 Trial

Immunotherapy with nivolumab as adjuvant therapy after CROSS regimen for patients with residual disease

Stage II/II Esophageal or GE junction cancers Adenocarcinoma or squamous cell

ChemoRT → Surgery *with residual disease on pathology*

Treatment Group: Nivolumab every 2 weeks x 4 months → every month x 8 months

*vs*

Control Group: No adjuvant therapy

⇒ Better survival in group with adjuvant nivolumab

(Kelly et al. 2021)

### Neoadjuvant Chemo for EsoCA

- MAGIC trial (gastric): ECF[1]→Surgery→ECF *vs* Surgery
- OEO2 Trial: (esophageal) Chemo→Surgery→ Chemo *vs* Surgery
- FLOT (gastric): FLOT[2]→Surgery→ FLOT *vs* ECF→Surgery→ECF
- EsoPEC: (esophageal):FLOT→Surgery→FLOT *vs* ChemoRT→Surgery (CROSS)

[1] Epirubicin, Cisplatin, 5FU
[2] 5FU, Leuvocorin, Oxaliplatin, Decetaxol

### OEO2 Clinical Trial

- 802 Esophageal adenocarcinoma and squamous cell
- Randomized to Chemo → Surgery → Chemo *vs* Surgery alone
- Chemotherapy with ECF (Epirubicin, Cisplatin, 5FU)
- 5-year survival 23% for chemo+surgery vs 17% for surgery (HR 0.84 p=0.03)

(Allum et al. 2009)

### Neo-Aegis Tral CROSS vs MAGIC/FLOT

- **Adenocarcinoma** T2-3 N0-3 M0 Tumor length <8cm
- ChemoRT arm: carboplatin + paclitaxel + 4140cGy
- Chemo arm: MAGIC (ECF) or FLOT (later in trial)
- No difference in overall survival
- R0 resection 96% with CROSS vs 82% with chemo
- pCR 12% with CROSS vs 4% with chemo

(**reynolds1015?**)

### Orientation Manual

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